



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

NAME – SUBASH G
REG NO – 192572150

COURSE NAME – OBJECT ORIENTED PROGRAMMING
COURSE CODE – DSA0108

CO2-AT2 -SCENARIO BASED

QUESTION 1 :

PROGRAM:

```
#include <iostream>
using namespace std;

int main()
{
    int choice;
    float baseCharge, specialistFee, serviceCharge, discount, total;

    cout << "==== Hospital Consultation Billing System =====" << endl;
    cout << "1. General Consultation" << endl;
    cout << "2. Specialist Consultation" << endl;

    cout << "\nEnter your choice: ";
    cin >> choice;

    cout << "Enter base consultation charge: ";
    cin >> baseCharge;

    if(choice == 2)
    {
        cout << "Enter specialist fee: ";
        cin >> specialistFee;
    }
    else
    {
        specialistFee = 0;
```

```

}

cout << "Enter service charge: ";
cin >> serviceCharge;

cout << "Enter discount amount (if any): ";
cin >> discount;

total = baseCharge + specialistFee + serviceCharge - discount;

cout << "\n----- Consultation Bill -----" << endl;
cout << "Patient Type: ";

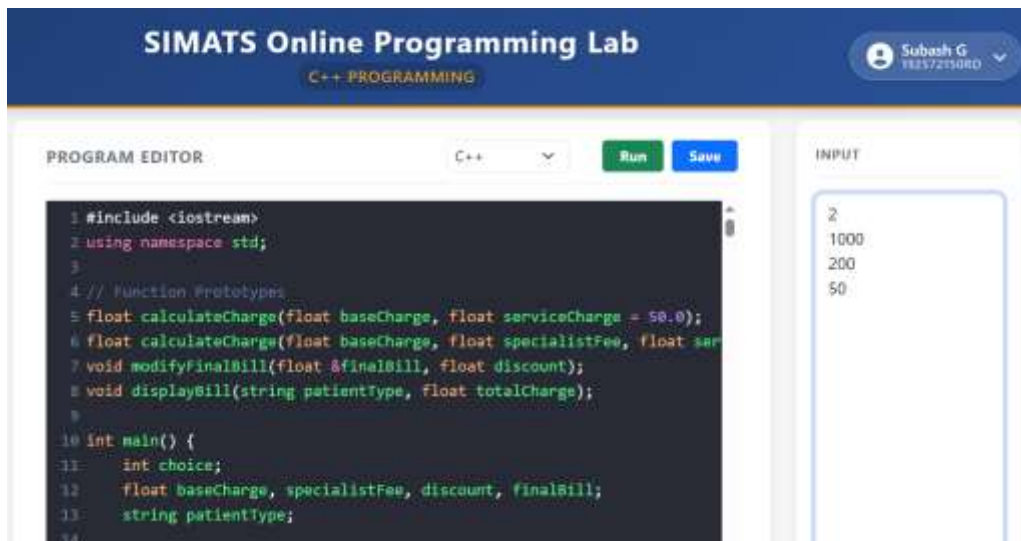
if(choice == 1)
    cout << "General Patient" << endl;
else if(choice == 2)
    cout << "Specialist Patient" << endl;
else
    cout << "Invalid Choice" << endl;

cout << "Total Consultation Charge: Rs. " << total << endl;

return 0;
}

```

SAMPLE INPUT:



The screenshot shows the SIMATS Online Programming Lab interface. The top header is blue with the text "SIMATS Online Programming Lab" and "C++ PROGRAMMING". A user profile icon for "Subash G" is visible in the top right. The main area is divided into two sections: "PROGRAM EDITOR" on the left and "INPUT" on the right. The "PROGRAM EDITOR" section contains a C++ code editor with the following code:

```

1 #include <iostream>
2 using namespace std;
3
4 // Function Prototypes
5 float calculateCharge(float baseCharge, float serviceCharge = 50.0);
6 float calculateCharge(float baseCharge, float specialistFee, float ser
7 void modifyFinalBill(float &finalBill, float discount);
8 void displayBill(string patientType, float totalCharge);
9
10 int main() {
11     int choice;
12     float baseCharge, specialistFee, discount, finalBill;
13     string patientType;
14

```

The "INPUT" section on the right contains a text area with the following input:

```

2
1000
200
50

```

OUTPUT:

OUTPUT

```
===== Hospital Consultation Charge System =====
1. General Consultation
2. Specialist Consultation
Enter your choice:
Enter base consultation charge:
Enter specialist fee:
Enter service charge:
Enter discount amount (if any):
```

```
----- Consultation Bill -----
Patient Type:
Specialist Patient
Total Consultation Charge: Rs. 1250
```

QUESTION 2 :

PROGRAM:

```
#include <iostream>
#include <string>
using namespace std;
```

```
struct Member
{
    int memberID;
    string name;
    string planType;
    int fee;
};
```

```
int main()
{
```

```
int n;
```

```
cout << "==== Fitness Centre Membership System =====> endl;
```

```
cout << "Enter number of members to register: ";
```

```
cin >> n;
```

```
Member m[n];
```

```
for(int i = 0; i < n; i++)
```

```
{
```

```
    cout << "\nEnter details for Member " << i + 1 << ":> endl;
```

```
    cout << "Member ID: ";
```

```
    cin >> m[i].memberID;
```

```
    cout << "Name: ";
```

```
    cin >> m[i].name;
```

```
    cout << "Plan Type (Basic/Standard/Premium): ";
```

```
    cin >> m[i].planType;
```

```
    if(m[i].planType == "Basic" || m[i].planType == "basic")
```

```
        m[i].fee = 500;
```

```
    else if(m[i].planType == "Standard" || m[i].planType == "standard")
```

```
        m[i].fee = 1000;
```

```
    else if(m[i].planType == "Premium" || m[i].planType == "premium")
```

```
        m[i].fee = 1500;
```

```
    else
```

```
        m[i].fee = 0;
```

```
}
```

```
cout << "\n==== All Membership Details =====> endl;
```

```
for(int i = 0; i < n; i++)
```

```
{
```

```
    cout << "\n----- Membership Details -----> endl;
```

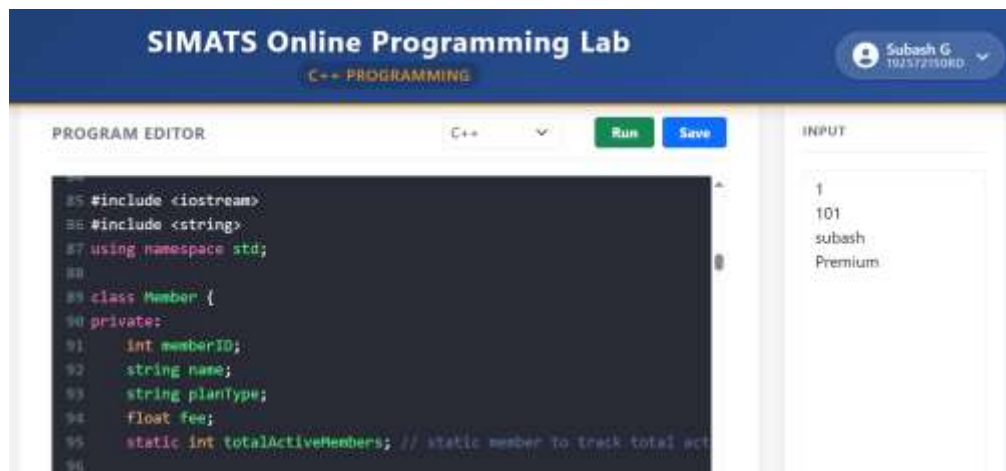
```

        cout << "Member ID  : " << m[i].memberID << endl;
        cout << "Name      : " << m[i].name << endl;
        cout << "Plan Type  : " << m[i].planType << endl;
        cout << "Fee        : Rs. " << m[i].fee << endl;
    }

    return 0;
}

```

SAMPLE INPUT:



OUTPUT:

OUTPUT

```

===== Fitness Centre Membership System =====
Enter number of members to register:
Enter details for Member 1:
Member ID: Name: Plan Type (Basic/Standard/Premium):
===== All Membership Details =====
----- Membership Details -----
Member ID    : 101
Name         : subash
Plan Type    : Premium
Fee          : Rs. 1500

```

QUESTION 3:

PROGRAM:

```
#include <iostream>

#include <iomanip>

#include <cmath>

using namespace std;

// Function to calculate EMI

double calculateEMI(double principal, double annualRate, int years)
{
    double monthlyRate = annualRate / (12 * 100);
    int months = years * 12;

    double emi = (principal * monthlyRate * pow(1 + monthlyRate, months)) /
        (pow(1 + monthlyRate, months) - 1);

    return emi;
}

int main()
{
    double principal, subsidy = 0, payableAmount;
    double interestRate = 10.0;
    char choice;
    int years;

    cout << "==== Loan Repayment System ====" << endl;
```

```

cout << "Enter Loan Amount (Principal): ";
cin >> principal;

cout << "Enter Loan Duration in Years: ";
cin >> years;


cout << "Do you want to enter interest rate? (Y/N): ";
cin >> choice;


if(choice == 'Y' || choice == 'y')
{
    cout << "Enter Annual Interest Rate (%): ";
    cin >> interestRate;
}

cout << "Enter Subsidy Amount (if any): ";
cin >> subsidy;

payableAmount = principal - subsidy;

double emi = calculateEMI(payableAmount, interestRate, years);
int months = years * 12;

double totalRepayment = emi * months;

double totalInterest = totalRepayment - payableAmount;

// Simple Interest (Reference)

double simpleInterest = (payableAmount * interestRate * years) / 100;

cout << fixed << setprecision(2);

cout << "\n===== Repayment Details =====" << endl;

cout << "Original Loan Amount: Rs. " << principal << endl;

```

```

cout << "Subsidy Provided:   Rs. " << subsidy << endl;

cout << "Payable Loan Amount: Rs. " << payableAmount << endl;

cout << "Annual Interest Rate: " << interestRate << " %" << endl;

cout << "Loan Duration:      " << years << " years (" << months << " months)" << endl;

cout << "\nMonthly EMI:       Rs. " << emi << endl;

cout << "Total Interest:      Rs. " << totalInterest << endl;

cout << "Total Repayment:     Rs. " << totalRepayment << endl;

cout << "\nSimple Interest (for reference): Rs. " << simpleInterest << endl;

return 0;

}

```

SAMPLE INPUT:

SIMATS Online Programming Lab
C++ PROGRAMMING
Subash G
192572150RD

PROGRAM EDITOR
C++
Run
Save

```

221  double rate;           // annual rate in %
222  double subsidy;       // subsidy amount
223  char choiceRate;
224
225  cout << "==== Loan Repayment System =====\n";
226  cout << "Enter Loan Amount (Principal): ";
227  cin >> loanAmount;
228
229  cout << "Enter Loan Duration in Years: ";
230  cin >> years;
231
232  cout << "Do you want to enter interest rate? (Y/N): ";
233  cin >> choiceRate;
234
235  // Default interest rate = 10% if user says N

```

INPUT
100000
5
N
5000

OUTPUT:

OUTPUT

```
===== Loan Repayment System =====  
Enter Loan Amount (Principal): Enter Loan Duration in Years: Do  
you want to enter interest rate? (Y/N): Enter Subsidy Amount (if  
any):  
===== Repayment Details =====  
Original Loan Amount: Rs. 100000.00  
Subsidy Provided:      Rs. 5000.00  
Payable Loan Amount:  Rs. 95000.00  
Annual Interest Rate: 10.00 %  
Loan Duration:        5 years (60 months)
```

OUTPUT

```
Payable Loan Amount:  Rs. 95000.00  
Annual Interest Rate: 10.00 %  
Loan Duration:        5 years (60 months)  
  
Monthly EMI:          Rs. 2018.47  
Total Interest:        Rs. 26108.15  
Total Repayment:       Rs. 121108.15  
  
Simple Interest (for reference): Rs. 47500.00
```

QUESTION 4:

PROGRAM:

```
#include <iostream>
```

```
#include <iomanip>
```

```
#include <string>
```

```
using namespace std;
```

```
class Parcel
```

```
{
```

public:

string type;

double weight;

double distance;

double packagingCharge;

double totalCharge;

void getDetails()

{

cout << "Enter Parcel Type (Normal/Express): ";

cin >> type;

cout << "Enter Weight (kg): ";

cin >> weight;

cout << "Enter Distance (km): ";

cin >> distance;

cout << "Enter Packaging Charge (or -1 to use default 50): ";

cin >> packagingCharge;

if(packagingCharge == -1)

packagingCharge = 50;

}

```

void calculateCharge()
{
    if(type == "Express" || type == "express")
    {
        totalCharge = (weight * distance * 6.25) + packagingCharge;
    }
    else
    {
        totalCharge = (weight * distance * 3.5) + packagingCharge;
    }
}

```

```

void display()
{
    cout << "\n----- Parcel Information -----" << endl;
    cout << "Parcel Type      : " << type << endl;
    cout << "Weight (kg)       : " << weight << endl;
    cout << "Distance (km)    : " << distance << endl;
    cout << "Packaging Charge : Rs. " << packagingCharge << endl;
    cout << fixed << setprecision(2);
    cout << "Total Delivery Charge (" << type << "): Rs. "
        << totalCharge << endl;
}
};

```

```
int main()
{
    int n;

    cout << "Enter number of parcels to process: ";
    cin >> n;

    Parcel p[n];

    for(int i = 0; i < n; i++)
    {
        cout << "\n--- Parcel " << i + 1 << " Details ---" << endl;

        p[i].getDetails();
        p[i].calculateCharge();
    }

    for(int i = 0; i < n; i++)
    {
        p[i].display();
    }

    return 0;
}
```

SAMPLE INPUT:

The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a blue header with the text "SIMATS Online Programming Lab" and "C++ PROGRAMMING". On the right, a user profile for "Subash G" with ID "192572150RD" is visible. Below the header, the interface is divided into two main sections: "PROGRAM EDITOR" and "INPUT".

In the "PROGRAM EDITOR" section, the following C++ code is visible:

```
367 }
368
369 void displayParcelInfo() const {
370     cout << "\n----- Parcel Information ----- \n";
371     cout << "Parcel Type      : " << parcelType << "\n";
372     cout << "Weight (kg)       : " << weightKg << "\n";
373     cout << "Distance (km)      : " << distanceKm << "\n";
374     cout << "Packaging Charge   : Rs. " << packagingCharge << "\n";
375
376     cout << fixed << setprecision(2);
377     if (parcelType == "Express" || parcelType == "EX" || parcelType == "EXP")
378         cout << "Total Delivery Charge (Express): Rs. "
```

In the "INPUT" section, the following input is shown:

```
1
Express
10
150
20
```

OUTPUT:

OUTPUT

```
Enter number of parcels to process:
--- Parcel 1 Details ---
Enter Parcel Type (Normal/Express): Enter Weight (kg): Enter
Distance (km): Enter Packaging Charge (or -1 to use default 50):
----- Parcel Information -----
Parcel Type      : Express
Weight (kg)      : 10
Distance (km)    : 150
Packaging Charge : Rs. 20
Total Delivery Charge (Express): Rs. 9400.00
```

QUESTION 5:

PROGRAM:

```
#include <iostream>
```

```
#include <iomanip>
```

```
#include <string>

using namespace std;

class WaterBill
{
private:
    int consumerID;
    string consumerType;
    double consumption;
    double baseBill;
    double finalBill;
    double serviceTax;
    double totalPayable;
public:
    void getDetails()
    {
        cout << "Enter Consumer ID: ";
        cin >> consumerID;
        cout << "Enter Consumer Type (Domestic/Commercial): ";
        cin >> consumerType;
        cout << "Enter Water Consumption (liters): ";
        cin >> consumption;
    }
    void calculateBill()
    {
```

```

// Rs.0.02 per liter
baseBill = consumption * 0.02;

// Subsidy
if(consumerType == "Domestic" || consumerType == "domestic")
    finalBill = baseBill * 0.80;    //20% subsidy
else
    finalBill = baseBill;

//10% Service Tax
serviceTax = finalBill * 0.10;

totalPayable = finalBill + serviceTax;
}

void displaySummary()
{
    cout << left
        << setw(12) << consumerType
        << setw(15) << consumption
        << "Subsidy-Adjusted Bill: Rs. "
        << fixed << setprecision(2)
        << finalBill << endl;
}

void displayDetails()
{

```

```
cout << fixed << setprecision(2);
```

```
cout << "\n===== Detailed Billing (Default Service Tax Applied) =====\n";
```

```
cout << "\n===== Billing Information =====\n";
```

```
cout << "Consumer ID      : " << consumerID << endl;
```

```
cout << "Consumer Type      : " << consumerType << endl;
```

```
cout << "Water Consumption    : " << consumption << " liters" << endl;
```

```
cout << "Base Bill Amount     : Rs. " << baseBill << endl;
```

```
cout << "Final Bill(after subsidy): Rs. " << finalBill << endl;
```

```
cout << "Service Tax (10.00%) : Rs. " << serviceTax << endl;
```

```
cout << "Total Payable       : Rs. " << totalPayable << endl;
```

```
}
```

```
};
```

```
int main()
```

```
{
```

```
int n;
```

```
cout << "Water Billing System\n\n";
```

```
cout << "Enter number of consumers to process: ";
```

```
cin >> n;
```



```
WaterBill consumer[n];
```

```
for(int i = 0; i < n; i++)
```

```
{
```

```
    cout << "\n--- Consumer " << i + 1 << " ---\n\n";
```

```
    consumer[i].getDetails();
```

```
    consumer[i].calculateBill();
```

```
}
```

```
// Summary
```

```
cout << "\n===== Consumer Summaries (After Subsidy) =====\n";
```

```
for(int i = 0; i < n; i++)
```

```
{
```

```
    consumer[i].displaySummary();
```

```
}
```

```
// Detailed Bills
```

```
for(int i = 0; i < n; i++)
```

```
{
```

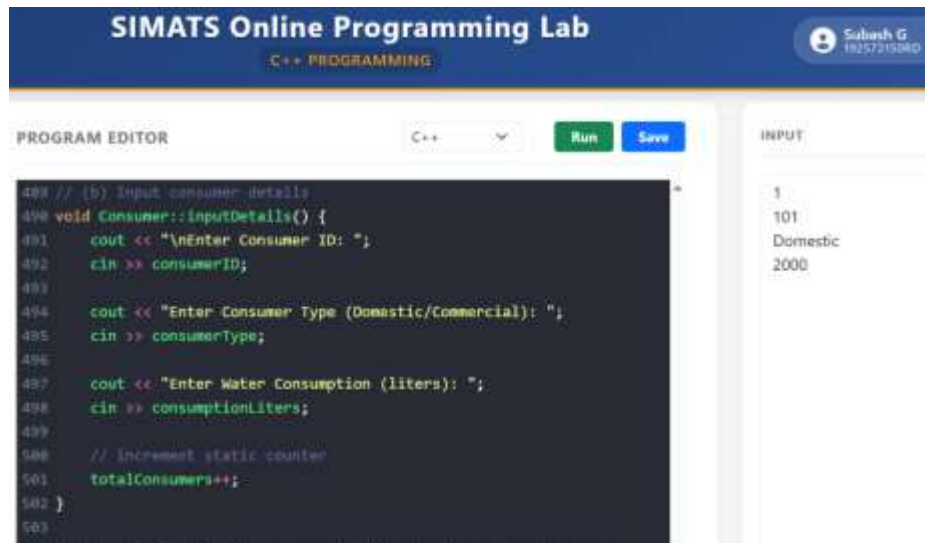
```
    consumer[i].displayDetails();
```

```
}
```

```
return 0;
```

```
}
```

SAMPLE INPUT:



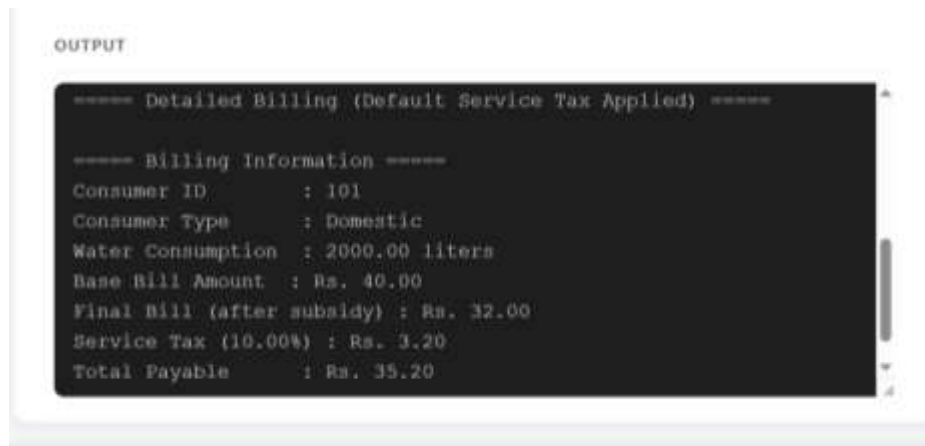
The screenshot shows the SIMATS Online Programming Lab interface. The top header is blue with the text "SIMATS Online Programming Lab" and "C++ PROGRAMMING". A user profile icon and name "Subash G" with ID "18257215280" are in the top right. The main area is divided into a "PROGRAM EDITOR" on the left and an "INPUT" section on the right. The editor shows C++ code for a consumer input function. The input section shows the values entered: 1, 101, Domestic, and 2000.

```
488 // (b) Input consumer details
489 void Consumer::inputDetails() {
490     cout << "\nEnter Consumer ID: ";
491     cin >> consumerID;
492
493     cout << "Enter Consumer Type (Domestic/Commercial): ";
494     cin >> consumerType;
495
496     cout << "Enter Water Consumption (liters): ";
497     cin >> consumptionLiters;
498
499     // Increment static counter
500     totalConsumers++;
501 }
502
503
```

INPUT

1
101
Domestic
2000

OUTPUT:



The screenshot shows the "OUTPUT" section of the SIMATS Online Programming Lab. It displays the output of the C++ program, which is a detailed billing statement. The output is formatted with asterisks for section headers and colon-separated key-value pairs for the billing information.

```
===== Detailed Billing (Default Service Tax Applied) =====

===== Billing Information =====
Consumer ID       : 101
Consumer Type     : Domestic
Water Consumption : 2000.00 liters
Base Bill Amount  : Rs. 40.00
Final Bill (after subsidy) : Rs. 32.00
Service Tax (10.00%) : Rs. 3.20
Total Payable     : Rs. 35.20
```